

Backend Modules Inventory

Below is a list of every module present in the backend API, mapped logically by domain. I detailed every endpoint, their controller functions, and explicitly flag their completion statuses (Completed, Partially Complete, Placeholder).

1. Auth Module (/v1/auth)

Status: Completed Handles all sign-ins, JWT lifecycle, and OTPs.

- POST /auth/request-otp - requestOtp (Completed: Triggers SMS/Email via Twilio/Resend).
- POST /auth/verify-otp - verifyOtp (Completed: Checks OTP, issues Access & Refresh Tokens).
- POST /auth/refresh - refresh (Completed: Robust token rotation with reuse-detection).
- POST /auth/logout - logout (Completed: Revokes tokens in database).
- POST /auth/google - googleSignIn (Completed: Integrates google-auth-library to auto-provision accounts).

2. User Module (/v1/users)

Status: Completed Handles the authenticated learners primary self-data points.

- GET /me - getMe (Completed: Returns current user contextual state).
- PATCH /me/profile - completeProfile (Completed: Zod-validated profile updates).
- PATCH /me/onboarding - completeOnboarding (Completed: Zod-validated onboarding configuration).

3. Curriculum Module (/v1/curriculum, /v1/units, etc.)

Status: Completed Handles read-only fetching of learning paths for standard users.

- GET /curriculum/standards - getStandards (Completed: Generic fetch).
- GET /curriculum/subjects - getSubjects (Completed: Specific to a standard ID or code).
- GET /units - getUnits (Completed: Filtered by Subject).
- GET /chapters - getChapters (Completed: Filtered by Unit).
- GET /lessons - getLessons (Completed: Complex implementation that merges lesson data with user's Attempt data to mark lessons as completed and unlocked).

- GET /lessons/:id -getLessonById (Completed).
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4. Attempt Module (/v1/attempts)

Status: Completed Handles quiz submissions and gamification triggers.

- POST /attempts/submit - submitAttempt (Completed: Extremely complex transactional endpoint. Evaluates quiz answers, calculates XP, levels up users, triggers wallet transactions for coins/diamonds, dynamically updates daily/weekly streaks and stats).
 - GET /attempts/quiz/:lessonId - getQuizForLesson (Completed: Safely redacts answer keys before sending the quiz payload to the frontend).
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5. Chat / AI Module (/v1/ai)

Status: Completed AI Tutor interface for users.

- POST /ai/chat - chat (Completed: Implements OpenAI contextual chatting. Includes robust rate-limiting via Redis/Memory, active cheat-detection, and dynamically queries the users past failed attempts to feed educational weakness context to the AI system prompt).
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6. Analytics Module (/v1/analytics)

Status: Completed Learner-facing statistical data.

- GET /analytics - getAnalytics (Completed: Deep MongoDB aggregations retrieving total lessons completed, average scores, longest streaks, moving 28-day XP history, and subject-wise strength breakdown).
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7. Leaderboard Module (/v1/leaderboards)

Status: Completed Global competition data.

- GET /leaderboards/weekly-growth - weeklyGrowth (Completed).
 - GET /leaderboards/mastery - mastery (Completed).
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8. Health Module (/v1/health, /v1/ready)

Status: Completed

- GET /health (Completed: Basic inline handler returning { status: 'ok' }).
 - GET /ready (Completed: Validates Mongoose connection state).
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9. Dashboard Module (/v1/dashboard)

Status: Partially Complete / Placeholders Intended to be the personalized home screen for the learner.

- GET /dashboard/home - getDashboardHome
 - Completed elements: Fetches user's current baseline profile, XP, level, streak, coins, and diamonds.
 - Placeholders / Incomplete elements: Hardcoded rank to null. Hardcoded continueLearning pointer to null. Returns empty arrays [] for leaderboard snippet and recentActivity.

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10. Admin Module (/v1/admin)

Status: Mostly Complete (1 Placeholder) Restricted to SuperAdmins and StandardAdmins. It contains multiple sub-controllers.

10.1 Admin Dashboard (admin.dashboard.controller)

- GET /metrics - getAdminDashboardMetrics (Completed: Generates system-wide usage widgets, quizzes taken, system uptime, and Agenda job stats).
- GET /retention - getRetentionAnalytics (Completed: Generates cohort retention numbers).

10.2 Admin Users (admin.users.controller)

- GET /users - listUsers (Completed).
- PATCH /users/:id - editUser (Completed).
- DELETE /users/:id - deleteUser (Completed).
- POST /users/:id/badges - awardBadge (Completed: Manual badge gifting).
- POST /users/:id/xp - awardXP (Completed: Manual XP gifting).
- POST /users/:id/reset-progress - resetUserProgress (Completed).
- GET /users/:id/profile - getUserProfile (Completed).
- POST /admins - createAdminAccount (Completed: Super-admin only).

10.3 Admin Curriculum (admin.controller)

- Contains 30+ endpoints for standard CRUD operations and soft-delete restorations for Standards, Subjects, Units, Chapters, and Lessons. (Completed)
- GET /quizzes/latest, POST /quizzes/version, PATCH /quizzes/:id/publish (Completed: Version-controlled quiz publishing).
- GET /audit - listAdminAuditLogs (Completed).

10.4 Admin Notifications (admin.notifications.controller)

- GET /notifications - listNotifications (Completed).
- POST /notifications - sendNotification (Completed: Integrates with OneSignal).

10.5 Admin Events (admin.events.controller)

- GET /events, POST /events, PATCH /events/:id, DELETE /events/:id (Completed).

10.6 Admin Badges (admin.badges.controller)

- GET /badges, POST /badges, PATCH /badges/:id, DELETE /badges/:id (Completed).

10.7 Admin Jobs (admin.jobs.controller)

- GET /jobs, POST /jobs/:id/retry, DELETE /jobs/:id (Completed: Direct interface to Agenda queues).

10.8 Admin System (admin.system.controller)

- GET /system/leaderboard, PATCH /system/leaderboard (Completed).
- POST /system/leaderboard/reset - resetLeaderboard (Completed).
- GET /system/api-logs - getApiLogs: Placeholder. Does not retrieve real pino logs or API gateway metrics; returns a hardcoded mockLogs array containing 4 fake requests.

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11. Additional Modules (/v1/....)